Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

Clin Chem. Author manuscript; available in PMC 2026 June 16.

Published in final edited form as:

Clin Chem. 2025 December 30; 72(1): 140–151. doi:10.1093/clinchem/hvaf154.

The Mutational Epidemiology of Childhood Cancer

Logan Spector[*], **Cassandra Clark**, **Zhanni Lu**, **Nathan Anderson**, **Erin Marcotte**, **Adam James de Smith**

University of Minnesota, Minneapolis, MN, United States

Abstract

Background: Childhood cancers comprise a variety of liquid and solid tumors that display different patterns of incidence than adult cancers. The majority have distinct molecular subtypes characterized by specific genomic driver events. The mutational processes that influence the somatic landscape of cancers also generate distinctive mutational signatures (mutSig). While these signatures are often inert, they do represent a fingerprint of the insult(s) present during mutagenesis. Associating mutSig with putatively causal exposures for pediatric cancer could inform future etiologic studies and elucidate the exposure pathways underlying risk.

Content: Here we review the epidemiology of pediatric cancers. We then summarize the knowledge around mutSig seen in pediatric cancer to date, discuss observed geographic and subtype-related variability, and discuss future efforts to characterize mutSig with unknown etiologies.

Summary: The diversity of childhood cancers and their molecular subtypes suggest etiologic heterogeneity. Detection of mutSig in childhood cancers has promoted hypothesis generation; e.g., the enrichment of UV-related signatures in aneuploid B-ALL has inspired new studies. Although the Mutographs projects were developed to investigate geographical variation in incidence, mutational epidemiology studies should also be employed to understand why certain mutSig are enriched in particular childhood cancers or subtypes. As pediatric cancers have lower mutational burdens than adult cancers, studying childhood cancer may also help determine the causes of mutSig with unknown etiologies. Given persistent differences in pediatric cancer risk by ancestry and socioeconomics, as well as the shifting global burden of childhood cancer, there is a need for studies with patients from diverse populations.

[*] spect012@umn.edu .

Author Contributions: The corresponding author takes full responsibility that all authors on this publication have met the following required criteria of eligibility for authorship: (a) significant contributions to the conception and design, acquisition of data, or analysis and interpretation of data; (b) drafting or revising the article for intellectual content; (c) final approval of the published article; and (d) agreement to be accountable for all aspects of the article thus ensuring that questions related to the accuracy or integrity of any part of the article are appropriately investigated and resolved. Nobody who qualifies for authorship has been omitted from the list.

Authors' Disclosures or Potential Conflicts of Interest: *No authors declared any potential conflicts of interest.*

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Keywords

Cancer; Pediatric Laboratory Medicine; Genetics

Descriptive Epidemiology of Childhood Cancer

Childhood cancers are a set of rare, histologically and molecularly heterogeneous malignant tumors defined by their age of occurrence. The age range is variably defined, typically as 0–14 or (as by the National Cancer Institute) 0–19 years at diagnosis. The spectrum of tumors in children contrasts sharply with that in adults. Whereas carcinomas are the overwhelming proportion of cancers in adults, they comprise a small minority of cancers in children. Instead, leukemia and lymphoma comprise nearly half of all pediatric tumors, central nervous system (CNS) tumors and sarcomas make up a substantial fraction, and a number of types such as the embryonal tumors are nearly exclusively diagnosed in children (Figure 1). The spectrum of tumors in children is so different that a separate International Classification of Childhood Cancer system, now in its third edition (ICCC-3), is the global standard for grouping pediatric tumors.(1)

Incidence of childhood cancer, and particular tumors, varies internationally. About 400,000 children are newly diagnosed each year according to one comprehensive, global analysis,(2) although there is a continuing need for improvements in estimation in underresourced areas without gold-standard cancer registration. In a series of publications using Cancer Incidence on Five Continents data from the International Agency for Research on Cancer, we found the widest differences in incidence across regions and countries for leukemias (3) and especially Ewing sarcoma,(4) while CNS tumors (5) and lymphomas (6) exhibited modest variation, and osteosarcoma relatively little.(7)

In the United States incidence is highest in children under 5 and is as high as 250 cases per million children in their first year of life and as low as 120 cases per million in mid-childhood (Figure 1).(8) However the age-incidence curves of individual tumors can be dramatically different. Acute lymphoblastic leukemia (ALL) displays a distinctive early-life peak in incidence between 2 and 5 years of age, while acute myeloid leukemia (AML) has steady incidence throughout childhood. The embryonal tumors (neuroblastoma, medulloblastoma, Wilm’s tumor, hepatoblastoma) each have highest incidence in infancy which declines to near zero after 5 years of age. The bone sarcomas (osteosarcoma and Ewing sarcoma) each show peak incidence roughly during puberty.

Childhood cancer incidence is patterned by sex and ancestry as well as age. Boys have greater incidence than girls for most cancers,(9) with the sex ratio ranging from approximately 1.1 (astrocytoma) to approximately 6 (Burkitt’s lymphoma). Moreover the sex ratio may change with age at diagnosis within types of cancer; for instance the ratio of males to females with ALL is approximately 1.1 in 1 year olds and rises to 2 in adolescents. Ancestry-related variation is similarly wide.(10) In the U.S. incidence of ALL is about 30% higher in Latino children than in White and Asian children, while Black children have less than 50% of the rate. For most other cancers, incidence is generally highest among White and Asian children, intermediate in Latino children, and far lower among Black children.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Ewing sarcoma is a notable outlier in its extremely wide ancestry-related differences in incidence between White, Asian, and especially Black children,(11) which recapitulates international observations.(4)

Etiology of Childhood Cancer

Analytic epidemiologic studies to understand causes of childhood cancer generally fall into three types: interview-based case-control studies, cohort and nested case-control studies formed through record-linkage, and molecular/genomic studies. Although more susceptible to selection and recall biases, interview-based casecontrol studies can examine potential risk factors which are not captured any other way (e.g. maternal diet during pregnancy). Conversely, record-linkage studies, while they may have rich medical data, usually cannot directly observe behavioral or lifestyle risk factors. Molecular/genomic studies include extensive tumor and germline sequencing or array genotyping, but to date have mostly come from clinical sources with minimal epidemiologic data. The Childhood Cancer and Leukemia International Consortium (CLIC: www.clic.ngo) aggregates most of the existing case-control data from around the globe thus examining tens of thousands of patients.(12) The International Childhood Cancer Cohort Consortium (I4C)(13) aggregates birth cohorts in order to enable highly-valuable, prospective investigations of childhood cancer etiology; however due to the rarity of cancer during childhood there are a small absolute number of cases. The amount of epidemiologic evidence for each cancer in children is roughly proportional to incidence.

Most types of childhood cancer are thought to initiate *in utero*.(14–17) Circumstantial evidence includes the young age at onset and histologic resemblance of tumor cells to fetal cognates. Direct evidence includes case reports of prenatal diagnosis for many solid tumors, while for leukemia there are both twin and “backtracking” studies. This has focused many etiologic studies on gestation and the perinatal period as windows of risk.

Structural birth defects of many types are strongly (OR 2–5) and very strongly (OR >5) associated with most types of childhood cancer.(18) Similarly, higher birth weight is associated with most childhood cancers with one exception - hepatoblastoma is for as-yet unknown reasons strongly inversely associated with birth weight.(19) Older parental age at birth is also associated with many cancers, to varying degrees (OR 1.05–1.2 per 5-years greater age).(20) Pathologies of pregnancy including pre-eclampsia, gestational diabetes, hyperemesis gravidarum, and obesity have each been associated with various cancers.(21,22) Maternal infections during pregnancy were associated with ALL and total cancer in childhood in the prospective I4C data.(23) More recently, the International Agency for Research on Cancer recently declared cytomegalovirus class 2b, “possibly carcinogenic in humans”, based mainly on studies examining congenital infection.(24)

Some environmental exposures during either pregnancy or childhood have been modestly but consistently associated with childhood cancers. These include exposures amenable to geographic information system (GIS) based analyses, air pollution,(25) heat,(26) light at night,(27) pesticides,(28) oenoculture,(29) and residential proximity to the oil and gas industry.(30) Therapeutic and, to a lesser degree, diagnostic ionizing radiation can

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

cause most childhood cancers,(31) although with falling levels of medical irradiation the exposure is mostly of historical interest. CLIC has conducted pooled analyses of parental occupational exposures to paint,(32) pesticides,(33) and electromagnetic fields(34) which appear associated with ALL. Behavioral risk factors such as parental smoking and alcohol use are, counter expectation, not notably associated with cancer in children.(35,36)

The genetic architecture(s) of childhood cancers are partially articulated. Rare, high penetrance variation identified by sequencing is strongly associated with most childhood cancers, but to varying degrees.(37) For most childhood cancers pathogenic/likely pathogenic (P/LP) variation in cancer susceptibility genes accounts for <10% of cases. Nearly all of some especially rare cancers, such as adrenocortical carcinoma, are due to P/LP genetic variants, as are nearly 30% of osteosarcomas.(38) The list of known childhood cancer susceptibility genes appears mostly filled-out but new ones are identified occasionally, most notably P/LP variants in *ELP1* in sonic-hedgehog mutated medulloblastoma.(39) Nearly alone among cancers, Ewing sarcoma has almost no P/LP associations; a comparatively weak association for P/LP variants (OR ~ 9) in *FANCC* was only described recently and has not yet been replicated.(40)

Common risk variants discovered by genomewide association studies (GWAS) are generally more strongly associated with childhood than adult cancers. Often the childhood cancer-associated risk alleles have effect sizes, or per-allele odds ratios, that are >1.5, which is far greater than usual in adults, in whom 1.1 is considered relatively strong. There are well-articulated common variant architectures for ALL(41) and neuroblastoma(42) among children of European descent from multiple GWAS including thousands of patients. As with GWAS in general, far fewer non-European patients have been genotyped. GWAS hits identified in European children have generally replicated transethnically in Black, Latino, and Asian patients when examined in ALL(43,44) and neuroblastoma.(45) In addition, two GWAS for ALL in Black and Latino children have identified ancestry-specific variants,(46,47) while a small GWAS among Japanese patients did not.(48) There are additionally single GWAS for Wilm's tumor,(49) osteosarcoma,(50) glioma,(51) medulloblastoma,(52) and most recently hepatoblastoma.(53) Lastly, Ewing sarcoma - an outlier cancer in many respects - is remarkably polygenic, with over a dozen loci identified in three phases of GWAS including only ~1,700 cases.(54) ALL and neuroblastoma both demonstrate "etiologic heterogeneity", meaning that SNP associations with molecularly-defined subtypes differ in degree. Lastly, it appears that trait-related variation may be avenues for future research, as there are fruitful integrative studies of blood traits in ALL(55) and transcriptome in hepatoblastoma.(53)

Molecular subtypes in childhood cancers

Childhood cancers comprise a variety of hematological malignancies (liquid tumors) and solid tumor types that display different patterns of incidence in relation to adult cancers, as discussed above. Furthermore, the majority of childhood cancers can be subdivided into distinct molecular subtypes that are characterized by specific genomic driver events. Liquid tumors in children are predominantly made up of acute leukemias, with ALL and AML comprising ~80% and 20% of childhood leukemias, respectively. Childhood ALL, which

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

develops from immature lymphocytes, can be subdivided into B-cell (B-ALL, approximately 85% of ALL) and T-cell (T-ALL, approximately 15%) ALL, and further heterogeneity exists within each of these immunophenotypes. The most common molecular subtypes of B-ALL are high hyperdiploidy, an aneuploidy with a characteristic pattern of chromosomal gains, and *ETV6::RUNX1* (formerly *TEL::AML1*) fusion, both of which occur in ~20% of ALL cases.(56) Other subtypes that have been detectable by cytogenetics for several decades include additional gene fusion events *TCF3::PBX1*, *BCR::ABL1*, and rearrangements of *KMT2A* (formerly *MLL*), as well as hypodiploid leukemias. Whole-genome sequencing (WGS) studies in recent years have discovered novel B-ALL subtypes including those driven by fusion events involving *DUX4*, *ZNF384*, or *MEF2D*, an array of rearrangements that underlie the *BCR::ABL1*-like (or Ph-like) subtype, and alterations of *PAX5*.(57,58) For T-ALL, most cases are driven by fusions involving *TAL1/2*, *TLX1*, *TLX3*, *HOXA*, *LMO1*/2, *LYL1*, and *NKX2*-1, and recent WGS of large numbers of cases has revealed a similarly heterogeneous set of driver events to B-ALL(59); indeed, at least 20 B-ALL subtypes and 15 T-ALL subtypes have been identified to date.(58,59)

While the incidence rate of ALL is much higher in children than in adults, the reverse is true for AML. However, the molecular subtypes of childhood AML have been well characterized, in particular to understand the relatively poor patient outcomes for this disease. While the French-American-British (FAB) classification system was traditionally used to divide AML (adult and childhood) into subtypes M0 through M7 based on leukemia cell morphology, the World Health Organization (WHO) classification based on specific genomic alterations is now more widely used.(60) Sequencing studies have revealed that the most frequent of these WHO subtypes in children include *NPM1*- and *CEBPA*-mutated AML and fusion events involving *RUNX1*, *CBFB*, and *KMT2A*.(61)

In childhood solid tumors, some cancer types can be subdivided into molecular subgroups that correlate with genomic alterations, similar to the hematological malignancies described above. For instance, neuroblastoma, which arises from immature nerve cells, can be stratified into low-, intermediate-, and high-risk groups that inform patient prognosis and treatment strategies.(62) Amplification of the *MYCN* gene occurs in ~25% of neuroblastoma patients and is a well-established biomarker of the high-risk subtype.(62,63) Among childhood brain tumors, medulloblastomas are classified into four main molecular subgroups, *WNT*, *SHH*, Group 3, and Group 4, that exhibit distinct epigenetic and transcriptional signatures and inform patient prognosis and treatment.(64,65) The *WNT* and *SHH* subgroups are driven by genomic alterations that activate the Wingless and Sonic hedgehog signaling pathways, respectively, while groups 3 and 4 are less well defined, although sequencing studies have identified mutations in a heterogeneous set of genes and suggest that additional subgroups of medulloblastoma may exist.(66) Among the sarcomas, rhabdomyosarcomas are primarily stratified into the embryonal and alveolar subtypes,(67) while osteosarcoma can be subdivided based on histology into osteoblastic, chondroblast, and fibroblastic groups(68) and potentially into further subgroups using multi-omics data.(69) In contrast to the cancers described above, Ewing sarcoma represents a more homogeneous disease that is driven by a specific translocation between the *EWSR1* and *FLI1* genes (*EWS::FLI1*) in ~90% of cases, with fusions between *EWSR1* and other partner genes in the remaining cases.(70)

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

The diverse nature of childhood cancers and their molecular subtypes suggests etiologic heterogeneity, and several instances of subtype-specific risk factors, particularly GWAS hits, have been described. For instance, a variant near *GATA3* is strongly associated with Ph-like ALL(71) while *BARD1* variants are associated with high risk neuroblastoma.(72) Rare P/LP variants in *ELP1* are specifically associated with medulloblastoma with *SHH* mutations.(73) Beyond genetics, the magnitude of association of birth weight with ALL varies by subtype(74) while there is widespread heterogeneity of associations with perinatal and familial risk characteristics between embryonal and alveolar rhabdomyosarcoma (the latter characterized by *PAX3::FOXO1* translocations).(75)

Signatures of Mutational Processes (mutSig) in Cancer

The driver mutations that characterize cancers are generally produced by some combination of DNA damage or modification and DNA repair or replication processes. These mutational processes generate distinctive mutational signatures,(76) which exist among “passenger” mutations. These signatures have variable etiologies and can be classified based upon standards established in the Catalogue of Somatic Mutations in Cancer (COSMIC) Mutational Signatures (Figure 2): single (SBS) and doublet (DBS) base substitutions, indels (ID), copy number (CN), structural variants (SV), and RNA single base substitutions (RNA SBS)(77). While the presence of particular signatures does not necessarily implicate a causal role for their associated exposures, they do represent a “fingerprint” of the insult(s), exogenous or endogenous, present during the mutagenic process. Associating mutational signatures with putatively causal exposures for pediatric cancer could inform future etiologic and causal studies and elucidate the exposure pathways underlying cancer risk. Such “mutational epidemiology” studies that integrate mutational signature analysis with epidemiological data to investigate cancer risk factors are already being conducted for adult cancers, as discussed below.

Tobacco smoking is a strong risk factor for many cancer types, and tobacco-specific mutational signatures have been identified.(78) In a multi-cancer study of 2,490 smokers and 1,062 non-smokers, smokers had significantly higher mutational burdens. Perhaps the most striking example of the differential mutational burden was seen in tobacco smokers diagnosed with lung adenocarcinoma. Compared to non-smokers, tobacco smokers diagnosed with lung adenocarcinoma had significantly elevated mutational burdens with respect to SBS2, 4, 5, and 13, and an elevated burden of SBS16. SBS4 was seen only in cancer types associated with smoking (e.g., lung squamous and adenocarcinoma, laryngeal; 10 of 17 total examined subtypes), and not seen in cancers of tissues not directly exposed to tobacco smoke (e.g., ovarian tissue). SBS4 is likely related to DNA misreplication as a result of damage by tobacco-associated carcinogens.(2) Notably, SBS4 was not prominent in either of two pediatric pan-cancer studies.(80,81)

Some similar patterns have been observed among never-smokers diagnosed with lung cancer.(82) Never-smokers reported to be exposed to secondhand smoke (“passive smoking”) had an 8% higher mutational burden than unexposed cases, but this increase was not evident in any particular mutational signature. Increasing air pollution was also associated with increased mutational burden, including a nearly 4-fold elevation of SBS4

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

(tobacco-associated) and significant elevation of clock-like SBS5. Notably, SBS40a was widespread (28% of total substitutions). SBS40a has been seen in kidney cancers including clear cell renal cell carcinomas;(83,84) its etiology is unknown.

A number of documented signatures have proposed etiologies. These proposed etiologies are generally based upon observational data and experimental models in humans, animals, cell lines, or yeast. (85,86) For example, SBS2 and SBS13 are thought to be related to activity by the AID/APOBEC family of cytidine deaminases due to their similarities with mutations caused by these enzymes in yeast models.(87) However, large proportions of signatures have not yet been assigned a proposed etiology. As of the time of writing, 22 of 67 (33%) documented SBS and 13 of 23 (57%) IDs have unknown etiologies in COSMIC v3.4. Efforts to characterize differences between signatures of known and unknown etiologies revealed that the latter varied in strand bias, G-C content, and nucleotide diversity, leading to hypotheses that some of these signatures may occur outside of gene regions.(86) Additionally, a number of mutations are thought to be sequencing artifacts and so are generally ignored; for instance, 9 documented SBS have been categorized as such.

Efforts to understand the etiologies of geographical variation in adult cancer incidence through mutational epidemiology have been spearheaded by the “Mutographs” project,(88) which has reported on significant regional differences in mutational signature burden and prevalence in multiple adult cancer types to date, including lung, kidney, and colorectal cancers.(82,83,89) For colorectal cancer, a higher burden of SBS88 (associated with the bacteria-produced mutagen colibactin)(90) was seen in countries with higher colorectal cancer incidence and enriched in early-onset disease.(89) In a study of clear cell renal cell carcinoma, an unknown signature was identified in 70% of Japanese cases and <2% of cases from other countries.(83) Conversely, mutational profiles were similar across esophageal squamous cell carcinoma cases from eight countries with varying incidence,(91) indicating that regional variation in mutational profile is likely highly cancer-specific. These results provide compelling evidence of international variability in mutational signature and burden, and of the potential factors underlying this variability, supporting a need for similar mutational epidemiology studies of childhood cancers.

MutSig in Childhood Cancer

Mutational burden in children is highly variable across tumor types.(80) Pediatric cancers exhibit fewer mutational signatures than adult cancers, and mutational burden is strongly correlated with age at diagnosis.(80,92) Our understanding of the frequency, distribution, and etiology of mutational signatures in pediatric cancers is limited. To understand the scope of the published peer-reviewed research on mutational signatures in pediatric cancer specifically, we conducted a search of the published literature using the query string TITLE-ABS (“mutational signature”) AND TITLE-ABS (pediatric) AND TITLE-ABS (cancer) OR TITLE-ABS (malignancy) OR TITLE-ABS (neoplasm) OR TITLE-ABS (tumor) OR TITLE-ABS (neuro) OR TITLE-ABS (leukemia) OR TITLE-ABS (blastoma) OR TITLE-ABS (lymphoma) OR TITLE-ABS (sarcoma) OR TITLE-ABS (myelo) in SCOPUS. The search yielded 36 results, which were title and abstract screened for relevance. Thirty articles

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

were then reviewed to identify those reporting individual-level mutSig data (Supplementary Table 1).

Eleven studies reported individual-level mutSig data for pediatric cases; most of these focused on single cancer subtypes.(58,80,81,93–100) For ease of comprehension we have visually represented these results together in Figure 3, with an alternate presentation in Supplementary Figure 1 and the distribution of relative contributions in Supplementary Figure 2; we caution that comparisons of mutSig across studies, and therefore their combination, are sensitive to the many analytic choices that produce these types of datasets. Thirty SBS signatures have been well-defined in adult cancers.(92)

Two pediatric pan-cancer studies, one in a North American (NA) (785 samples, 27 subtypes) cohort and one in a European (EU) cohort (879 samples, 24 subtypes), are responsible for much of the knowledge in this area to date.(80,81) Of the 30 mutSig observed in adult populations,(92) 29 were observed in the NA cohort and 17 in the EU cohort. Those signatures that appear in at least two cancer subtypes in at least one of the cohorts are shown in Table 1. Of pediatric solid tumors, astrocytomas, ependymomas, and retinoblastoma had among the lowest median SBS burden per mB, while osteo- and rhabdomyosarcoma, Burkitt lymphoma, glioma, and neuroblastoma had among the highest. Hematologic malignancies, including B-ALL and AML, tended to have low to moderate mutational burdens as compared to other subtypes save for T-ALL, which had a mutational burden comparable to neuroblastoma. The relative mutational burden across cancer subtypes was similar between the two cohorts. Notably, hepatoblastoma had the lowest mutational burden in the EU cohort; this cancer was however not reported in the NA cohort.

Common signatures seen in studies of pediatric cancer included those associated with homologous recombination repair defects or late replication errors(101) and clock-like signatures associated with age at diagnosis, seen in both adult and pediatric cancers.(102) A few signatures are putatively associated with environmental exposures; e.g., signatures of UV exposure (SBS7a/b and ID13) that have been seen only in acute lymphoblastic leukemias and embryonal tumors.(80) Notably, SBS3 and 8 (associated with DNA repair and replication) appeared in 30–40% more cancer subtypes in the EU cohort, while SBS5 and 18 were 77% and 42% more common in the NA cohort. SBS16 was seen in 54% of cancer subtypes in the EU cohort but in 0% of the NA cohort.

While the Mutographs projects were developed to investigate the potential etiologic factors underlying geographical variation in cancer incidence, mutational epidemiology studies should also be implemented to understand why certain mutational signatures are enriched in particular childhood cancers or molecular subtypes of childhood cancers. A striking example of this is the enrichment of UV-related mutational signatures in aneuploid subtypes of childhood B-ALL (hyperdiploidy and iAMP21), with UV-induced mutations appearing to occur subsequent to chromosomal gains.(58) It is not clear why preleukemic lymphocytes with chromosomal aneuploidies may be more prone to UV exposure, but it does suggest that UV radiation exposure in early life may be a risk factor for childhood ALL. Indeed, increasing UV radiation exposure was recently associated with an increased risk of childhood ALL,(103) and future studies that leverage information on both mutational

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

signatures and UV exposure data among childhood ALL cases are warranted. Enrichment of the APOBEC-related signatures SBS2 and 13 in childhood ALL cases specifically with *ETV6::RUNX1* fusion or the *ETV6::RUNX1*-like phenotype(58) points to possible subtype-specific effects of early-life infections or immune dysregulation. Another mutational signature enriched in childhood ALL cases is SBS18, which is linked to DNA damage caused by reactive oxygen species (ROS). The ROS signature has been detected in many childhood solid tumor types, including neuroblastoma, rhabdomyosarcoma, hepatoblastoma, Ewing sarcoma, and medulloblastoma,(80) and epidemiological studies should explore potential associations with exposure to air pollution and other environmental toxicants that can influence ROS metabolism. Other notable observations that warrant further investigation include detection of the APOBEC signatures in osteosarcoma,(80,95) nitrosamine exposure-related signature SBS29 in hepatoblastoma patients in Brazil,(104) and several signatures of unknown etiology that have been found in multiple childhood cancer types including AML, neuroblastoma, and Wilms tumor.(95)

Conclusion

As whole genome sequencing is increasingly incorporated into clinical tumor genetic workflows,(105) laboratories may in time come to be expected to call and annotate mutSig in addition to driver mutations. Multiple tools have been developed for the detection of mutSig in tumor sequencing data,(106,107) including methods to identify the specific COSMIC signatures in individual patient samples.(108,109) There are multiple reasons to assess mutSig for clinical purposes, most importantly there is emerging evidence that some mutational signatures in particular cancers may predict survival above and beyond established prognostic factors.(110) Secondly, as epidemiologic and laboratory studies refine the known etiologies of mutSig and resolve those with unknown etiologies, it seems possible that it will be feasible to determine the etiology of individual tumors, which may be tracked for public health purposes. In addition, given that several mutSig are attributed to chemotherapy exposure, where there is doubt the signatures may be used to determine whether samples are pre- or post-therapy.

Prior to incorporating mutSig into routine clinical practice, however, there are clear needs for additional research into their import in childhood cancers. There are several initiatives funded by the National Institutes of Health which have produced (or are continuing to do so) large-scale tumor and germline sequencing. They include the Molecular Profiling to Predict Response to Therapy (MP2PRT) studies of ALL(111) and Wilms' tumor,(112) the Molecular Characterization Initiative (MCI)(113) examining solid tumors with poor prognosis, and the Gabriella Miller Kids First Program (GMKF)(114) of multiple tumors. All of these efforts draw patients primarily from the Children's Oncology Group (COG)(115) and therefore may be linked to more extensive clinical, demographic, and epidemiologic data including address-derived exposures. As COG includes institutions across the United States, Canada, Australia, and New Zealand patients reside in highly variable areas with regards to climate and environmental exposures, thus the pieces are already in place to produce mutograph-style studies or mutational epidemiology investigations of pediatric cancer. However, given the large and persistent differences in risk of pediatric cancers by ancestry(116) and economic development,(3) as well as the shifting

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

global burden of childhood cancer,(117) the need for studies with patients from diverse low and middle income countries seems clear.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Nonstandard Abbreviations:

Human Genes:

ELP1	elongator acetyltransferase complex subunit 1
FANCC	FA complementation group C
ETV6	ETS variant transcription factor 6
RUNX1	RUNX family transcription factor 1
TCF3	transcription factor 3
PBX1	PBX homeobox 1
BCR	BCR activator of RhoGEF and GTPase
ABL1	ABL proto-oncogene 1, non-receptor tyrosine kinase
KMT2A	lysine methyltransferase 2A
DUX4	double homeobox 4
ZNF384	zinc finger protein 384
MEF2D	myocyte enhancer factor 2D
PAX5	paired box 5
TAL1	TAL bHLH transcription factor 1, erythroid differentiation factor
TAL2	TAL bHLH transcription factor 2
TLX1	T cell leukemia homeobox 1
TLX3	T cell leukemia homeobox 3
HOXA	homeobox A cluster
LMO1	LIM domain only 1
LMO2	LIM domain only 2
LYL1	LYL1 basic helix-loop-helix family member
NKX2–1	NK2 homeobox 1
NPM1	nucleophosmin 1

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

CEBPA	CCAAT enhancer binding protein alpha
CBFB	core-binding factor subunit beta
MYCN	MYCN proto-oncogene, bHLH transcription factor
WNT	Wnt family
SHH	sonic hedgehog signaling molecule
EWSR1	EWS RNA binding protein 1
FLI1	Fli-1 proto-oncogene
ETS	transcription factor
GATA3	GATA binding protein 3
BARD1	BRCA1 associated RING domain 1
PAX3	paired box 3
FOXO1	forkhead box O1

References

1. Steliarova-Foucher E, Stiller C, Lacour B, Kaatsch P. International Classification of Childhood Cancer, third edition. Cancer. 2005;103:1457–67. [PubMed: 15712273]
2. Ward ZJ, Yeh JM, Bhakta N, Frazier AL, Atun R. Estimating the total incidence of global childhood cancer: a simulation-based analysis. Lancet Oncol. 2019;20:483–93. [PubMed: 30824204]
3. Hubbard AK, Spector LG, Fortuna G, Marcotte EL, Poynter JN. Trends in International Incidence of Pediatric Cancers in Children Under 5 Years of Age: 1988–2012. JNCI Cancer Spectr. 2019;3:pkz007. [PubMed: 30984908]
4. Spector LG, Hubbard AK, Diessner BJ, Machiela MJ, Webber BR, Schiffman JD. Comparative international incidence of Ewing sarcoma 1988 to 2012. Int J Cancer. 2021;149:1054–66. [PubMed: 33961701]
5. Williams LA, Hubbard AK, Scheurer ME, Spector LG, Poynter JN. Trends in paediatric central nervous system tumour incidence by global region from 1988 to 2012. Int J Epidemiol. 2021;50:116–27. [PubMed: 33221912]
6. Chun GYC, Sample J, Hubbard AK, Spector LG, Williams LA. Trends in pediatric lymphoma incidence by global region, age and sex from 1988–2012. Cancer Epidemiol. 2021;73:101965. [PubMed: 34174724]
7. Rojas GA, Hubbard AK, Diessner BJ, Ribeiro KB, Spector LG. International trends in incidence of osteosarcoma (1988–2012). Int J Cancer. 2021;149:1044–53. [PubMed: 33963769]
8. National Cancer Institute. National Childhood Cancer Registry Explorer (NCCR*Explorer). https://nccrexplorer.ccdi.cancer.gov/ (Accessed July 2025)
9. Williams LA, Richardson M, Marcotte EL, Poynter JN, Spector LG. Sex ratio among childhood cancers by single year of age. Pediatr Blood Cancer. 2019;66:e27620. [PubMed: 30815990]
10. Marcotte EL, Domingues AM, Sample JM, Richardson MR, Spector LG. Racial and ethnic disparities in pediatric cancer incidence among children and young adults in the United States by single year of age. Cancer. 2021;127:3651–63. [PubMed: 34151418]
11. Diessner BJ, Weigel BJ, Murugan P, Zhang L, Poynter JN, Spector LG. Racial and Ethnic Differences in Sarcoma Incidence Are Independent of Census-Tract Socioeconomic Status. Cancer Epidemiol Biomarkers Prev. 2020;29:2141–8. [PubMed: 32928933]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

12. Metayer C, Milne E, Clavel J, Infante-Rivard C, Petridou E, Taylor M, et al. The Childhood Leukemia International Consortium. Cancer Epidemiol. 2013;37:336–47. [PubMed: 23403126]
13. Tikellis G, Dwyer T, Paltiel O, Phillips GS, Lemeshow S, Golding J, et al. The International Childhood Cancer Cohort Consortium (I4C): A research platform of prospective cohorts for studying the aetiology of childhood cancers. Paediatr Perinat Epidemiol. 2018;32:568–83. [PubMed: 30466188]
14. Marshall GM, Carter DR, Cheung BB, Liu T, Mateos MK, Meyerowitz JG, et al. The prenatal origins of cancer. Nat Rev Cancer. 2014;14:277–89. [PubMed: 24599217]
15. de Smith AJ, Spector LG. In Utero Origins of Acute Leukemia in Children. Biomedicines. 2024;12:236. [PubMed: 38275407]
16. Okonechnikov K, Joshi P, Sepp M, Leiss K, Sarropoulos I, Murat F, et al. Mapping pediatric brain tumors to their origins in the developing cerebellum. Neuro-Oncol. 2023;25:1895–909. [PubMed: 37534924]
17. Kristensen DM, Sonne SB, Ottesen AM, Perrett RM, Nielsen JE, Almstrup K, et al. Origin of pluripotent germ cell tumours: the role of microenvironment during embryonic development. Mol Cell Endocrinol. 2008;288:111–8. [PubMed: 18420341]
18. Lupo PJ, Schraw JM, Desrosiers TA, Nembhard WN, Langlois PH, Canfield MA, et al. Association Between Birth Defects and Cancer Risk Among Children and Adolescents in a Population-Based Assessment of 10 Million Live Births. JAMA Oncol. 2019;5:1150–8. [PubMed: 31219523]
19. O'Neill KA, Murphy MF, Bunch KJ, Puumala SE, Carozza SE, Chow EJ, et al. Infant birthweight and risk of childhood cancer: international population-based case control studies of 40 000 cases. Int J Epidemiol. 2015;44:153–68. [PubMed: 25626438]
20. Domingues A, Moore KJ, Sample J, Kharoud H, Marcotte EL, Spector LG. Parental Age and Childhood Lymphoma and Solid Tumor Risk: A Literature Review and Meta-Analysis. JNCI Cancer Spectr. 2022;6:pkac040. [PubMed: 35639955]
21. Marley AR, Domingues A, Ghosh T, Turcotte LM, Spector LG. Maternal Body Mass Index, Diabetes, and Gestational Weight Gain and Risk for Pediatric Cancer in Offspring: A Systematic Review and Meta-Analysis. JNCI Cancer Spectr. 2022;6:pkac020. [PubMed: 35603850]
22. Yin C, Adenekan TA, Deng C, Hansen J, Iwundu CN, Heck JE. Application of an obstetric comorbidity index to predict childhood cancer risk: a population based case-control study in Denmark. Br J Cancer. Nature Publishing Group; 2025;1–9.
23. He J-R, Hirst JE, Tikellis G, Phillips GS, Ramakrishnan R, Paltiel O, et al. Common maternal infections during pregnancy and childhood leukaemia in the offspring: findings from six international birth cohorts. Int J Epidemiol. 2021;51:769–77.
24. Karagas MR, Kaldor J, Michaelis M, Muchengeti MM, Alfaiate D, Argirion I, et al. Carcinogenicity of hepatitis D virus, human cytomegalovirus, and Merkel cell polyomavirus. Lancet Oncol. 2025;S1470–2045(25)00403-6.
25. Clark CJ, Haynes D, Lu Z, Sample JM, McGuinn LA, Hoang TT, et al. Ambient Fine Particulate Matter, Residential Greenness, and Childhood Cancer Risk by Trimester of Exposure in Minnesota 2000–2014: A Case-control Study. Cancer Epidemiol Biomark Prev. 2025; 34(10):1730–1739.
26. Rogne T, Wang R, Wang P, Deziel NC, Metayer C, Wiemels JL, et al. High ambient temperature in pregnancy and risk of childhood acute lymphoblastic leukaemia: an observational study. Lancet Planet Health. Elsevier; 2024;8:e506–14. [PubMed: 38969477]
27. Zhong C, Wang R, Morimoto LM, Longcore T, Franklin M, Rogne T, et al. Outdoor artificial light at night, air pollution, and risk of childhood acute lymphoblastic leukemia in the California Linkage Study of Early-Onset Cancers. Sci Rep. Nature Publishing Group; 2023;13:583. [PubMed: 36631468]
28. Lombardi C, Thompson S, Ritz B, Cockburn M, Heck JE. Residential proximity to pesticide application as a risk factor for childhood central nervous system tumors. Environ Res. 2021;197:111078. [PubMed: 33798513]
29. Mancini M, Hémon D, de Crouy-Chanel P, Guldner L, Faure L, Clavel J, et al. Association between Residential Proximity to Viticultural Areas and Childhood Acute Leukemia Risk

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

in Mainland France: GEOCAP Case-Control Study, 2006–2013. Environ Health Perspect. Environmental Health Perspectives; 2023;131:107008. [PubMed: 37850750]

30. Clark CJ, Johnson NP, Soriano M, Warren JL, Sorrentino KM, Kadan-Lottick NS, et al. Unconventional Oil and Gas Development Exposure and Risk of Childhood Acute Lymphoblastic Leukemia: A Case–Control Study in Pennsylvania, 2009–2017. Environ Health Perspect. Environmental Health Perspectives; 2022;130:087001. [PubMed: 35975995]
31. Wakeford R, Doll R, Bithell JF. The risk of childhood cancer from low doses of ionising radiation received in utero. In: Low Doses of Ionizing Radiation: Biological Effects and Regulatory Control: IAEA-TECDO. International Atomic Energy Agency; 1997. page 391–4. https://research.manchester.ac.uk/en/publications/the-risk-of-childhood-cancer-from-low-doses-of-ionising-radiation (Accessed July 2025)
32. Bailey HD, Fritschi L, Metayer C, Infante-Rivard C, Magnani C, Petridou E, et al. Parental occupational paint exposure and risk of childhood leukemia in the offspring: findings from the Childhood Leukemia International Consortium. Cancer Causes Control CCC. 2014;25:1351–67. [PubMed: 25088805]
33. Bailey HD, Fritschi L, Infante-Rivard C, Glass DC, Miligi L, Dockerty JD, et al. Parental occupational pesticide exposure and the risk of childhood leukemia in the offspring: findings from the childhood leukemia international consortium. Int J Cancer. 2014;135:2157–72. [PubMed: 24700406]
34. Talibov M, Olsson A, Bailey H, Erdmann F, Metayer C, Magnani C, et al. Parental occupational exposure to low-frequency magnetic fields and risk of leukaemia in the offspring: findings from the Childhood Leukaemia International Consortium (CLIC). Occup Environ Med. 2019;76:746–53. [PubMed: 31358566]
35. Rumrich IK, Viluksela M, Vähäkangas K, Gissler M, Surcel H-M, Hänninen O. Maternal Smoking and the Risk of Cancer in Early Life – A Meta-Analysis. PLoS ONE. 2016;11:e0165040. [PubMed: 27824869]
36. Hu Z, Ye J, Shi S, Luo C, Wang T, Liu Y, et al. Maternal smoking, consumption of alcohol, and caffeinated beverages during pregnancy and the risk of childhood brain tumors: a meta-analysis of observational studies. BMC Public Health. 2024;24:1238. [PubMed: 38711042]
37. Kratz CP. Re-envisioning genetic predisposition to childhood and adolescent cancers. Nat Rev Cancer. 2025;25:109–28. [PubMed: 39627375]
38. Mirabello L, Zhu B, Koster R, Karlins E, Dean M, Yeager M, et al. Frequency of Pathogenic Germline Variants in Cancer-Susceptibility Genes in Patients With Osteosarcoma. JAMA Oncol. 2020;6:724–34. [PubMed: 32191290]
39. Waszak SM, Robinson GW, Gudenas BL, Smith KS, Forget A, Kojic M, et al. Germline Elongator mutations in Sonic Hedgehog medulloblastoma. Nature. Nature Publishing Group; 2020;580:396–401.
40. Gillani R, Camp SY, Han S, Jones JK, Chu H, O'Brien S, et al. Germline predisposition to pediatric Ewing sarcoma is characterized by inherited pathogenic variants in DNA damage repair genes. Am J Hum Genet. 2022;109:1026–37. [PubMed: 35512711]
41. de Smith AJ, Jiménez-Morales S, Mejía-Aranguré JM. The genetic risk of acute lymphoblastic leukemia and its implications for children of Latin American origin. Front Oncol. 2024;13:1299355. [PubMed: 38264740]
42. Ritenour LE, Randall MP, Bosse KR, Diskin SJ. Genetic susceptibility to neuroblastoma: current knowledge and future directions. Cell Tissue Res. 2018;372:287–307. [PubMed: 29589100]
43. Perez-Andreu V, Xu H, Yang JJ. The novel susceptibility variants for childhood acute lymphoblastic leukemia. J Natl Cancer Inst. 2013;105:1512–3.
44. Urayama KY, Takagi M, Kawaguchi T, Matsuo K, Tanaka Y, Ayukawa Y, et al. Regional evaluation of childhood acute lymphoblastic leukemia genetic susceptibility loci among Japanese. Sci Rep. 2018;8:789. [PubMed: 29335448]
45. Latorre V, Diskin SJ, Diamond MA, Zhang H, Hakonarson H, Maris JM, et al. Replication of neuroblastoma SNP association at the BARD1 locus in African-Americans. Cancer Epidemiol Biomark Prev Publ Am Assoc Cancer Res Cosponsored Am Soc Prev Oncol. 2012;21:658–63.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

46. Im C, Raduski A, Mills LJ, Johnson RA, DeWan A, Ma X, et al. Abstract 2845: Novel genetic risk loci for B-cell acute lymphoblastic leukemia in African American children: Findings from the Admixture and Risk of Acute Leukemia (ADMIRAL) Study. Cancer Res. 2024;84:2845.
47. de Smith AJ, Wahlster L, Jeon S, Kachuri L, Black S, Langie J, et al. A noncoding regulatory variant in IKZF1 increases acute lymphoblastic leukemia risk in Hispanic/Latino children. Cell Genomics. 2024;4:100526. [PubMed: 38537633]
48. Hangai M, Kawaguchi T, Takagi M, Matsuo K, Jeon S, Chiang CWK, et al. Genome-wide assessment of genetic risk loci for childhood acute lymphoblastic leukemia in Japanese patients. Haematologica. 2024;109:1247–52. [PubMed: 37881853]
49. Turnbull C, Perdeaux ER, Pernet D, Naranjo A, Renwick A, Seal S, et al. A genome-wide association study identifies susceptibility loci for Wilms tumor. Nat Genet. 2012;44:681–4. [PubMed: 22544364]
50. Savage SA, Mirabello L, Wang Z, Gastier-Foster JM, Gorlick R, Khanna C, et al. Genome-wide association study identifies two susceptibility loci for osteosarcoma. Nat Genet. 2013;45:799–803. [PubMed: 23727862]
51. Foss-Skiftesvik J, Li S, Rosenbaum A, Hagen CM, Stoltze UK, Ljungqvist S, et al. Multi-ancestry genome-wide association study of 4069 children with glioma identifies 9p21.3 risk locus. Neuro-Oncol. 2023;25:1709–20. [PubMed: 36810956]
52. Dahlin AM, Wibom C, Andersson U, Bybjerg-Grauholm J, Deltour I, Hougaard DM, et al. A genome-wide association study on medulloblastoma. J Neurooncol. 2020;147:309–15. [PubMed: 32056145]
53. Xie T, Sorenson JC, Spector LG, Pankratz N, Huang RS, Hiyama E, et al. Multi-ancestry transcriptome-wide association study identifies candidate genes associated with hepatoblastoma. Cancer Epidemiol Biomark Prev. 2025;34(8):1405–1414.
54. Machiela MJ, Grünewald TGP, Surdez D, Reynaud S, Mirabeau O, Karlins E, et al. Genome-wide association study identifies multiple new loci associated with Ewing sarcoma susceptibility. Nat Commun. 2018;9:3184. [PubMed: 30093639]
55. Kachuri L, Jeon S, DeWan AT, Metayer C, Ma X, Witte JS, et al. Genetic determinants of blood-cell traits influence susceptibility to childhood acute lymphoblastic leukemia. Am J Hum Genet. 2021;108:1823–35. [PubMed: 34469753]
56. Pui C-H, Nichols KE, Yang JJ. Somatic and germline genomics in paediatric acute lymphoblastic leukaemia. Nat Rev Clin Oncol. 2019;16:227–40. [PubMed: 30546053]
57. Gu Z, Churchman ML, Roberts KG, Moore I, Zhou X, Nakitandwe J, et al. PAX5-driven subtypes of B-progenitor acute lymphoblastic leukemia. Nat Genet. 2019;51:296–307. [PubMed: 30643249]
58. Brady SW, Roberts KG, Gu Z, Shi L, Pounds S, Pei D, et al. The genomic landscape of pediatric acute lymphoblastic leukemia. Nat Genet. 2022;54:1376–89. [PubMed: 36050548]
59. Pölönen P, Di Giacomo D, Seffernick AE, Elsayed A, Kimura S, Benini F, et al. The genomic basis of childhood T-lineage acute lymphoblastic leukaemia. Nature. 2024;632:1082–91. [PubMed: 39143224]
60. Khoury JD, Solary E, Abla O, Akkari Y, Alaggio R, Apperley JF, et al. The 5th edition of the World Health Organization Classification of Haematolymphoid Tumours: Myeloid and Histiocytic/Dendritic Neoplasms. Leukemia. 2022;36:1703–19. [PubMed: 35732831]
61. Bolouri H, Farrar JE, Triche T, Ries RE, Lim EL, Alonzo TA, et al. The molecular landscape of pediatric acute myeloid leukemia reveals recurrent structural alterations and age-specific mutational interactions. Nat Med. 2018;24:103–12. [PubMed: 29227476]
62. Matthay KK, Maris JM, Schleiermacher G, Nakagawara A, Mackall CL, Diller L, et al. Neuroblastoma. Nat Rev Dis Primer. 2016;2:16078.
63. Huang M, Weiss WA. Neuroblastoma and MYCN. Cold Spring Harb Perspect Med. 2013;3:a014415. [PubMed: 24086065]
64. Northcott PA, Korshunov A, Pfister SM, Taylor MD. The clinical implications of medulloblastoma subgroups. Nat Rev Neurol. 2012;8:340–51. [PubMed: 22565209]
65. Northcott PA, Robinson GW, Kratz CP, Mabbott DJ, Pomeroy SL, Clifford SC, et al. Medulloblastoma. Nat Rev Dis Primer. 2019;5:11.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

66. Northcott PA, Buchhalter I, Morrissy AS, Hovestadt V, Weischenfeldt J, Ehrenberger T, et al. The whole-genome landscape of medulloblastoma subtypes. Nature. 2017;547:311–7. [PubMed: 28726821]

67. Skapek SX, Ferrari A, Gupta AA, Lupo PJ, Butler E, Shipley J, et al. Rhabdomyosarcoma. Nat Rev Dis Primer. 2019;5:1.

68. Beird HC, Bielack SS, Flanagan AM, Gill J, Heymann D, Janeway KA, et al. Osteosarcoma. Nat Rev Dis Primer. 2022;8:77.

69. Jiang Y, Wang J, Sun M, Zuo D, Wang H, Shen J, et al. Multi-omics analysis identifies osteosarcoma subtypes with distinct prognosis indicating stratified treatment. Nat Commun. 2022;13:7207. [PubMed: 36418292]

70. Lessnick SL, Ladanyi M. Molecular pathogenesis of Ewing sarcoma: new therapeutic and transcriptional targets. Annu Rev Pathol. 2012;7:145–59. [PubMed: 21942527]

71. Williams LA, Yang JJ, Hirsch BA, Marcotte EL, Spector LG. Is There Etiologic Heterogeneity between Subtypes of Childhood Acute Lymphoblastic Leukemia? A Review of Variation in Risk by Subtype. Cancer Epidemiol Biomark Prev. 2019;28:846–56.

72. Capasso M, Devoto M, Hou C, Asgharzadeh S, Glessner JT, Attiyeh EF, et al. Common variations in BARD1 influence susceptibility to high-risk neuroblastoma. Nat Genet. Nature Publishing Group; 2009;41:718–23. [PubMed: 19412175]

73. Waszak SM, Robinson GW, Gudenas BL, Smith KS, Forget A, Kojic M, et al. Germline Elongator mutations in Sonic Hedgehog medulloblastoma. Nature. Nature Publishing Group; 2020;580:396–401. [PubMed: 32296180]

74. O'Neill KA, Bunch KJ, Vincent TJ, Spector LG, Moorman AV, Murphy MFG. Immunophenotype and cytogenetic characteristics in the relationship between birth weight and childhood leukemia. Pediatr Blood Cancer. 2012;58:7–11. [PubMed: 21681930]

75. Lupo PJ, Luna-Gierke RE, Chambers TM, Tavelin B, Scheurer ME, Melin B, et al. Perinatal and familial risk factors for soft tissue sarcomas in childhood through young adulthood: A population-based assessment in 4 million live births. Int J Cancer. 2020;146:791–802. [PubMed: 30980537]

76. Alexandrov LB, Nik-Zainal S, Wedge DC, Aparicio SAJR, Behjati S, Biankin AV, et al. Signatures of mutational processes in human cancer. Nature. 2013;500:415–21. [PubMed: 23945592]

77. COSMIC. Mutational Signatures. Available from: https://cancer.sanger.ac.uk/signatures/ (Accessed January 2025)

78. Alexandrov LB, Ju YS, Haase K, Van Loo P, Martincorena I, Nik-Zainal S, et al. Mutational signatures associated with tobacco smoking in human cancer. Science. 2016;354:618–22. [PubMed: 27811275]

79. Nik-Zainal S, Kucab JE, Morganella S, Glodzik D, Alexandrov LB, Arlt VM, et al. The genome as a record of environmental exposure. Mutagenesis. 2015;30:763–70. [PubMed: 26443852]

80. Thatikonda V, Islam SMA, Autry RJ, Jones BC, Gröbner SN, Warsow G, et al. Comprehensive analysis of mutational signatures reveals distinct patterns and molecular processes across 27 pediatric cancers. Nat Cancer. 2023;4:276–89. [PubMed: 36702933]

81. Gröbner SN, Worst BC, Weischenfeldt J, Buchhalter I, Kleinheinz K, Rudneva VA, et al. The landscape of genomic alterations across childhood cancers. Nature. 2018;555:321–7. [PubMed: 29489754]

82. Díaz-Gay M, Zhang T, Hoang PH, Khandekar A, Zhao W, Steele CD, et al. The mutagenic forces shaping the genomic landscape of lung cancer in never smokers. medRxiv. 2024;2024.05.15.24307318.

83. Senkin S, Moody S, Díaz-Gay M, Abedi-Ardekani B, Cattiaux T, Ferreiro-Iglesias A, et al. Geographic variation of mutagenic exposures in kidney cancer genomes. Nature. 2024;629:910–8. [PubMed: 38693263]

84. Scelo G, Riazalhosseini Y, Greger L, Letourneau L, Gonzàlez-Porta M, Wozniak MB, et al. Variation in genomic landscape of clear cell renal cell carcinoma across Europe. Nat Commun. Nature Publishing Group; 2014;5:5135. [PubMed: 25351205]

85. Zou X, Owusu M, Harris R, Jackson SP, Loizou JI, Nik-Zainal S. Validating the concept of mutational signatures with isogenic cell models. Nat Commun. 2018;9:1744. [PubMed: 29717121]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

86. Hu X, Xu Z, De S. Characteristics of mutational signatures of unknown etiology. NAR Cancer. 2020;2:zcaa026. [PubMed: 33015626]

87. Chan K, Roberts SA, Klimczak LJ, Sterling JF, Saini N, Malc EP, et al. An APOBEC3A hypermutation signature is distinguishable from the signature of background mutagenesis by APOBEC3B in human cancers. Nat Genet. Nature Publishing Group; 2015;47:1067–72.

88. Stratton MR, Humphreys L, Alexandrov LB, Balmain A, Brennan P, Campbell PJ, et al. Implementing Mutational Epidemiology on a Global Scale: Lessons from Mutographs. Cancer Discov. 2025;15:22–7. [PubMed: 39801236]

89. Díaz-Gay M, dos Santos W, Moody S, Kazachkova M, Abbasi A, Steele CD, et al. Geographic and age variations in mutational processes in colorectal cancer. Nature. Nature Publishing Group; 2025;1–2.

90. Pleguezuelos-Manzano C, Puschhof J, Rosendahl Huber A, van Hoeck A, Wood HM, Nomburg J, et al. Mutational signature in colorectal cancer caused by genotoxic pks+ E. coli. Nature. Nature Publishing Group; 2020;580:269–73. [PubMed: 32106218]

91. Moody S, Senkin S, Islam SMA, Wang J, Nasrollahzadeh D, Cortez Cardoso Penha R, et al. Mutational signatures in esophageal squamous cell carcinoma from eight countries with varying incidence. Nat Genet. Nature Publishing Group; 2021;53:1553–63. [PubMed: 34663923]

92. Alexandrov LB, Kim J, Haradhvala NJ, Huang MN, Tian Ng AW, Wu Y, et al. The repertoire of mutational signatures in human cancer. Nature. 2020;578:94–101. [PubMed: 32025018]

93. van der Ham CG, Suurenbroek LC, Kleisman MM, Antić Ž, Lelieveld SH, Yeong M, et al. Mutational mechanisms in multiply relapsed pediatric acute lymphoblastic leukemia. Leukemia. 2024;38:2366–75. [PubMed: 39232206]

94. Guerrini-Rousseau L, Merlevede J, Denizeau P, Andreiuolo F, Varlet P, Puget S, et al. Glioma oncogenesis in the Constitutional mismatch repair deficiency (CMMRD) syndrome. Neuro-Oncol Adv. 2024;6:vdae120.

95. Ma X, Liu Y, Liu Y, Alexandrov LB, Edmonson MN, Gawad C, et al. Pan-cancer genome and transcriptome analyses of 1,699 paediatric leukaemias and solid tumours. Nature. 2018;555:371–6. [PubMed: 29489755]

96. Byron SA, Hendricks WPD, Nagulapally AB, Kraveka JM, Ferguson WS, Brown VI, et al. Genomic and Transcriptomic Analysis of Relapsed and Refractory Childhood Solid Tumors Reveals a Diverse Molecular Landscape and Mechanisms of Immune Evasion. Cancer Res. 2021;81:5818–32. [PubMed: 34610968]

97. Lambo S, Gröbner SN, Rausch T, Waszak SM, Schmidt C, Gorthi A, et al. The Molecular Landscape of ETMR at Diagnosis and Relapse. Nature. 2019;576:274–80. [PubMed: 31802000]

98. Newman S, Nakitandwe J, Kesserwan CA, Azzato EM, Wheeler DA, Rusch M, et al. Genomes for Kids: The Scope of Pathogenic Mutations in Pediatric Cancer Revealed by Comprehensive DNA and RNA Sequencing. Cancer Discov. 2021;11:3008–27. [PubMed: 34301788]

99. Brady SW, Liu Y, Ma X, Gout AM, Hagiwara K, Zhou X, et al. Pan-neuroblastoma analysis reveals age- and signature-associated driver alterations. Nat Commun. Nature Publishing Group; 2020;11:5183. [PubMed: 33056981]

100. Hirsch TZ, Pilet J, Morcrette G, Roehrig A, Monteiro BJ, Molina L, et al. Integrated genomic analysis identifies driver genes and cisplatin-resistant progenitor phenotype in pediatric liver cancer. Cancer Discov. 2021;11:2524–43. [PubMed: 33893148]

101. Singh VK, Rastogi A, Hu X, Wang Y, De S. Mutational signature SBS8 predominantly arises due to late replication errors in cancer. Commun Biol. 2020;3:421. [PubMed: 32747711]

102. Alexandrov LB, Jones PH, Wedge DC, Sale JE, Campbell PJ, Nik-Zainal S, et al. Clock-like mutational processes in human somatic cells. Nat Genet. 2015;47:1402–7. [PubMed: 26551669]

103. Little MP, Mai JZ, Fang M, Chernyavskiy P, Kennerley V, Cahoon EK, et al. Solar ultraviolet radiation exposure, and incidence of childhood acute lymphocytic leukaemia and non-Hodgkin lymphoma in a US population-based dataset. Br J Cancer. 2024;130:1441–52. [PubMed: 38424165]

104. Aguiar TFM, Rivas MP, Costa S, Maschietto M, Rodrigues T, Sobral de Barros J, et al. Insights Into the Somatic Mutation Burden of Hepatoblastomas From Brazilian Patients. Front Oncol. 2020;10:556. [PubMed: 32432034]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

105. Cuppen E, Elemento O, Rosenquist R, Nikic S, IJzerman M, Zaleski ID, et al. Implementation of Whole-Genome and Transcriptome Sequencing Into Clinical Cancer Care. JCO Precis Oncol. Wolters Kluwer; 2022;e2200245.
106. Koh G, Degasperi A, Zou X, Momen S, Nik-Zainal S. Mutational signatures: emerging concepts, caveats and clinical applications. Nat Rev Cancer. 2021;21:619–37. [PubMed: 34316057]
107. Islam SMA, Díaz-Gay M, Wu Y, Barnes M, Vangara R, Bergstrom EN, et al. Uncovering novel mutational signatures by de novo extraction with SigProfilerExtractor. Cell Genomics. 2022;2:None.
108. Medo M, Ng CKY, Medová M. A comprehensive comparison of tools for fitting mutational signatures. Nat Commun. 2024;15:9467. [PubMed: 39487150]
109. Díaz-Gay M, Vangara R, Barnes M, Wang X, Islam SMA, Vermes I, et al. Assigning mutational signatures to individual samples and individual somatic mutations with SigProfilerAssignment. Bioinforma Oxf Engl. 2023;39:btad756.
110. Karihtala P, Kilpivaara O, Porvari K. Mutational signatures and their association with cancer survival and gene expression in multiple cancer types. Int J Cancer. 2025;156:114–29. [PubMed: 39194330]
111. Chang T-C, Chen W, Qu C, Cheng Z, Hedges D, Elsayed A, et al. Genomic Determinants of Outcome in Acute Lymphoblastic Leukemia. J Clin Oncol. Wolters Kluwer; 2024;42:3491–503.
112. Gadd S, Huff V, Skol AD, Renfro LA, Fernandez CV, Mullen EA, et al. Genetic changes associated with relapse in favorable histology Wilms tumor: A Children's Oncology Group AREN03B2 study. Cell Rep Med. 2022;3:100644. [PubMed: 35617957]
113. Jagu S, Guidry Auvil JM, Reaman G. The childhood cancer data initiative: enabling data sharing to drive research advances and transform pediatric cancer diagnosis and treatment. Curr Opin Pediatr. 2025;37:42–7. [PubMed: 39699099]
114. Hudson A, Fournier M, Coulombe J, Daee D. Using existing pediatric cancer data from the Gabriella Miller Kids First Data Resource Program. JNCI Cancer Spectr. 2023;7:pkad079. [PubMed: 37788089]
115. Lupo PJ, Marcotte EL, Scheurer ME, Poynter JN, Spector LG, COG Epidemiology Committee. Children's Oncology Group's 2023 blueprint for research: Epidemiology. Pediatr Blood Cancer. 2023;70 Suppl 6:e30566. [PubMed: 37449937]
116. Marcotte EL, Domingues AM, Sample JM, Richardson MR, Spector LG. Racial and ethnic disparities in pediatric cancer incidence among children and young adults in the United States by single year of age. Cancer. 2021;127:3651–63. [PubMed: 34151418]
117. The global burden of childhood and adolescent cancer in 2017: an analysis of the Global Burden of Disease Study 2017. Lancet Oncol. 2019;20:1211–25. [PubMed: 31371206]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

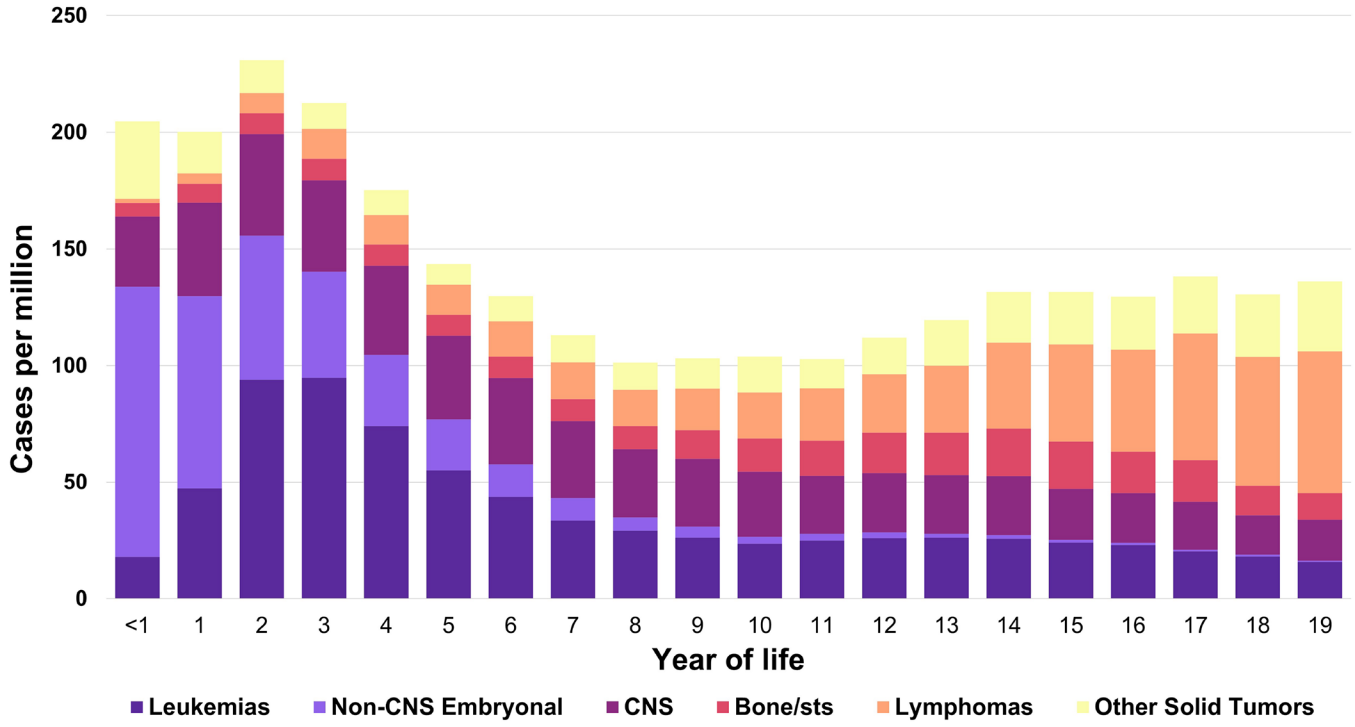


Figure 1:

Distribution of incidence of tumors across the pediatric age spectrum, SEER 21, 2013–2022. Sts, soft tissue sarcoma.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

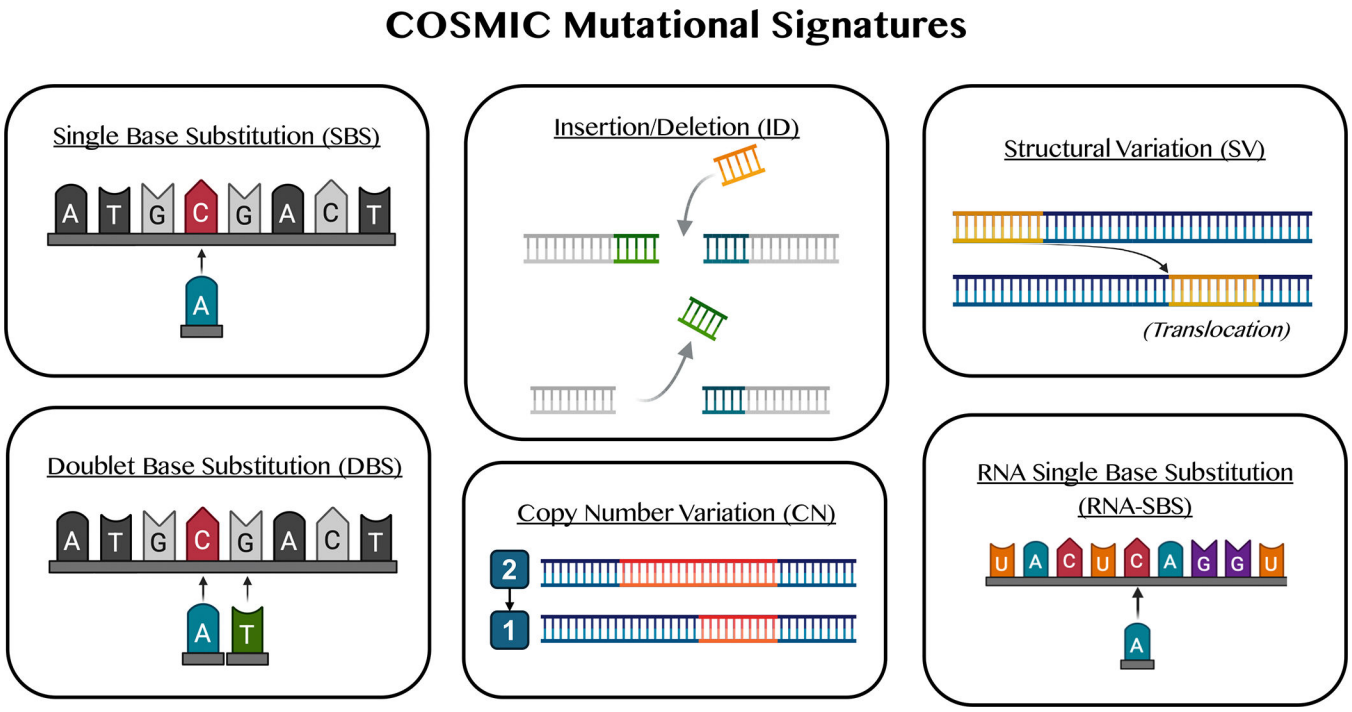


Figure 2:
Visual representation of the Catalogue of Somatic Mutations in Cancer (COSMIC) Mutational Signatures.

Author Manuscript
Author Manuscript
Author Manuscript
Author Manuscript



B-ALL= B-cell acute lymphoblastic leukemia, T-ALL= T-cell acute lymphoblastic leukemia, AML= acute myeloid leukemia, BL= burkitt lymphoma, DLBCL= diffuse large B-cell lymphoma, HGG= high grade glioma, LGG= low grade glioma, EP= ependymoma, PA= pilocytic astrocytoma, MB= medulloblastoma, ATRT= atypical teratoid rhabdoid tumor, ETMR= embryonal tumor with multilayered rosettes, PB= pineoblastoma, CP= craniopharyngioma, NB= neuroblastoma, RB= retinoblastoma, WT= wilms tumor, RCC= renal cell carcinoma, HB= hepatoblastoma, HCC= hepatocellular carcinoma, OS= osteosarcoma, ES= Ewing sarcoma, RMS= rhabdomyosarcoma, MPNST= malignant peripheral nerve sheath tumor, SS= synovial sarcoma, DSRCT= desmoplastic small round cell tumor, ACC= adrenocortical carcinoma, TC= thyroid carcinoma

Figure 3:

Percentage of cases within each cancer type with a detected mutational signature and its mean relative contribution, based on publicly available individual-level data (n = 3,707). Only signatures found in >5% of cases in at least one cancer type are shown. Individual-level data from 11 studies (58,80,81,93–100) with varying signature-calling methods were merged for this analysis.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 1:

Common mutational signatures observed in two pediatric pan-cancer surveys and probable etiologies.

Mutational signature	Europe[a]	North America[b]	Probable etiology
	N (%)[c] of subtypes	N (%)[c] of subtypes	
SBS1	23 (96)	27 (100)	Clock-like deamination of 5-methylcytosine
SBS3	11 (46)	4 (15)	BRCA/DSB repairs
SBS5	8 (33)	27 (100)	Clock-like
SBS8	21 (88)	12 (44)	Homologous recombination repair defects or late replication errors; associated with genomic instability
SBS16	13 (54)	0 (0)	Unknown
SBS18	5 (21)	17 (63)	ROS-related DNA damage
[d] SBS P1[a] or SBS31, SBS35[b]	2 (8)	3, 4 (11, 14)	Platinum treatment

Signatures appearing in at least two cancer subtypes in at least one of the cohorts are included.

[a] Gröbner et al., 2018 (81)

[b] Thatikonda et al., 2023 (80)